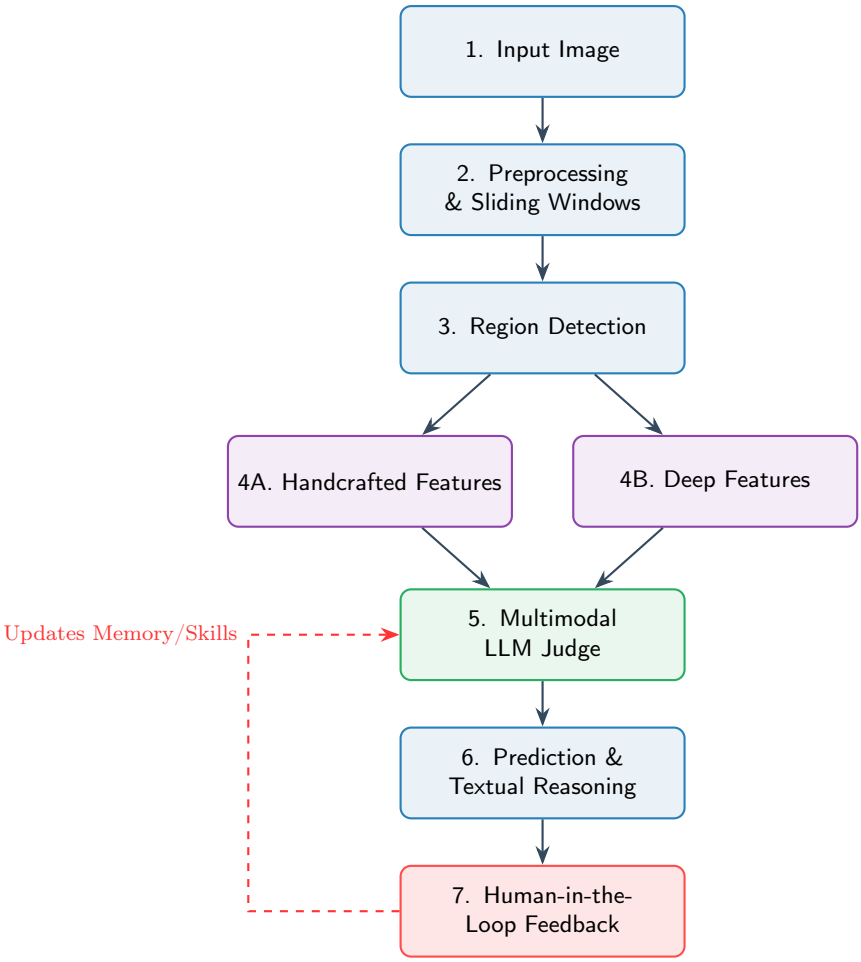
Feature Identification for Art Authentication

Final Architecture Detailed Breakdown

System Goal

Analyze ragamala paintings intelligently using sliding-window detection, multimodal understanding, machine learning, and adaptive reasoning to achieve 85% - 90% accuracy with continuous improvement.



Detailed Architecture Workflow

1. INPUT PAINTING IMAGE

1.1 User Upload: User uploads a painting image (JPG / PNG / WEBP).

1.2 Validate File: Check file type, size, resolution, corrupted file.

1.3 Store Image: Save original image in secure storage.

1.4 Generate ID: Assign a unique image ID for tracking.

1.5 Load Image: Load image into memory for processing.

2. PREPROCESSING – PRESERVE ORIGINAL RESOLUTION + SLIDING WINDOW PREPARATION

2.1 Keep Original (No Global Resize): Preserve original resolution to retain fine details.

2.2 Denoise: Apply noise reduction filter (Median / NL-Means).

2.3 Color Normalize: Normalize color balance and white balance.

2.4 Lighting Adjust: Adjust brightness, contrast, gamma, exposure.

2.5 Sharpen Image: Enhance edges and fine details (Unsharp Mask).

2.6 Format Convert: Convert to RGB format (if not already).

2.7 Prepare Sliding Windows (Overlap): Divide image into overlapping windows.

Sliding Window Configuration:

- **Window Size ($w \times h$):** e.g., 512×512 pixels
- **Stride / Step Size (s):** e.g., 256 pixels (50% overlap)
- **Overlap:** (typically 30% – 70%)
- **Padding (if required):** Handle border regions
- **Output:** Set of overlapping windows covering the entire image

3. OBJECT / REGION DETECTION USING SLIDING WINDOWS

3.1 Run Detection on Each Window: Apply object/region detection model (YOLO / DETR etc.) on every window.

3.2 Merge Detections: Combine overlapping detections from multiple windows (Non-Max Suppression).

3.3 Refine Detections: Remove duplicates, filter by confidence threshold.

3.4 Classify Regions (Preliminary): Assign coarse labels (e.g., human, animal, architecture, plant, object, text).

3.5 Crop Relevant Regions: Crop detected regions from original image (using coordinates).

3.6 Store Regions: Store cropped regions with metadata (type, bbox, confidence).

4A. HANDCRAFTED FEATURE EXTRACTION (ON REGIONS)

4A.1 Color (HSV): Hue, Saturation, Value histograms.

4A.2 Texture: GLCM, LBP, Gabor filters.

4A.3 Stroke: Stroke length, direction, morphology.

4A.4 Shape: Contours, shapes, geometry.

4A.5 Composition: Rule of thirds, symmetry, focal points, layout analysis.

4B. DEEP FEATURE EXTRACTION (ON CROPPED REGIONS)

4B.1 CNN Backbone: Extract deep features using ResNet / VGG / EfficientNet.

4B.2 Feature Vector: Generate fixed-size feature vector for each region.

4B.3 Region Aggregation: Aggregate region features (Attention / Mean Pooling).

4B.4 Global Feature Vector: Combine aggregated region features into a global image representation.

5. MULTIMODAL LLM AGENT / "JUDGE"

5.1 Input to Agent: Image (global + region features) + metadata + history to LLM.

5.2 Visual Understanding: LLM analyzes visual elements (colors, strokes, textures, composition).

5.3 Context Reasoning: LLM reasons about period, genre, style, mood, time, season, cultural context.

5.4 Knowledge Integration: LLM integrates art history knowledge & training data.

5.5 Skill Memory Consultation: LLM consults agent memory for similar past cases and outcomes.

5B. REASONING & DECISION MAKING

5.6 Decision Making: LLM makes final judgments (multi-label predictions).

5.7 Confidence Estimation: LLM estimates confidence (Score 0 – 100%).

5.8 Explanation Generation: LLM generates natural language explanation for predictions.

6. PREDICTION OUTPUT + TEXTUAL REASONING (EVALUATION PIPELINE)

6.1 Artist / Style Prediction: Predicted artist or style.

6.2 Period / Genre Prediction: Predicted period and genre.

6.3 Quality / Authenticity Scoring: Quality score and authenticity assessment.

6.4 Confidence Score: Overall confidence score (85% – 90%).

6.5 Explanation / Reasoning: Detailed textual reasoning for each prediction.

6.6 Key Visual Highlights: Highlight important regions / elements in the painting.

6.7 Output Packaging: Pack results in structured JSON / API response.

7. HUMAN-IN-THE-LOOP FEEDBACK ENGINE

7.1 Human Review: Expert reviews the predictions and reasoning.

7.2 Provide Feedback: Human provides corrections / confirmations.

7.3 Feedback Storage: Store feedback in agent memory system.

7.4 Update Skill File: Update agent skill file with new rules / patterns.

7.5 Continuous Learning: Agent learns and improves from new feedback.

7.6 Model Fine-tune (Optional): Fine-tune ML models or LLM prompts based on feedback.

7.7 Close Feedback Loop: Feedback loop feeds back to memory and models.